

# SARATHI AI (CASCADE INDIA)

## SIH Prototype Blueprint & Teammate Building Guide

Telegram Task Triage · UPI Escrow Hold · Worker Web Portal · AI Automated Proof Verification

**Executive Vision:** Sarathi AI brings the exact architecture of **Cascade** to India. Users communicate strictly over **Telegram** (or WhatsApp). Informational queries are answered instantly by AI for **0**. Tasks requiring real human labor (store calls, app testing, local errands, CA/legal advice) generate a price quote. Upon user acceptance, funds are locked in a **Dummy UPI Escrow**. When a worker submits proof on the **Worker Web Dashboard**, an **AI Proof Verification Engine** automatically inspects the proof against task requirements. Once verified, escrow funds are instantly released to the worker's UPI ID.

### 1. End-to-End Judge Presentation Demo Flow

To win at Smart India Hackathon (SIH), present this exact 6-step live workflow to the judges:

| Step | Actor        | Action / Screen                                                                                                                                                                                                                                             | System Mechanics & Status |
|------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| 1    | User (Phone) | Sends task in Telegram: <i>'Call Om Stationary Kora Triage Engine classifies task as 'Gig' (Network). Calculates quote for 120 minutes at 1000 INR/hr = 12000 INR. Escrow holds 12000 INR. Bot sends quote card with 'Approve & Pre-Auth 12000' button.</i> |                           |
| 2    | Telegram Bot | Bot replies with Quote Card: <b>'Price: 12000 (Held in UPI Escrow)'</b> × <b>'Approve & Pre-Auth 12000'</b></code>                                                                                                                                          |                           |
| 3    | User (Phone) | Taps <b>'Approve & Pre-Auth 12000'</b> button in Telegram. Telegram Pre-Auth Escrow Locked!</b> Status: <code>ESCROW_HOLD</code>                                                                                                                            |                           |
| 4    | Worker (Web) | Worker opens <code>/worker</code> portal, views task, and clicks <b>'Claim Task'</b>. Status transitions to <code>IN_PROGRESS</code>. Assigned to worker.                                                                                                   |                           |
| 5    | Worker (Web) | Worker completes task & uploads proof: <i>'Spoke with Om Stationary Kora. Status transitions to <code>AVAILABLE @ 12000 INR</code>. AI Verification Engine initiated.</i>                                                                                   |                           |
| 6    | AI & Engine  | <b>AI Proof Verification Engine</b> validates proof against requirements. AI confirms <b>'Proof Verified'</b>. Status transitions to <code>SETTLED</code>. 12000 released to worker's UPI ID.                                                               |                           |

### 2. System Architecture & Component Responsibilities

| Module Name                 | Tech Stack                 | Key Responsibilities                                                                                                             |
|-----------------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| 1. Cascade Landing Page     | HTML5, CSS3, JS (Vanilla)  | Serves marketing UI at <code>/</code> matching Cascade aesthetic. Features Hero section, About, Contact, and Team.               |
| 2. Worker Web Dashboard     | HTML5, JS, REST APIs       | Standalone portal at <code>/worker</code> for gig workers/experts to view escrow-backed tasks, submit proof, and track earnings. |
| 3. Backend API Server       | Python (Flask / FastAPI)   | Central engine handling state persistence, worker feeds, escrow Virtual Account ledger, & Telegram bot bridge.                   |
| 4. Telegram Bot Bridge      | python-telegram-bot / HTTP | Polls Telegram API, sends Markdown quote cards with inline buttons, & delivers completion notifications.                         |
| 5. AI Triage & Proof Engine | Google Gemini / Groq API   | 1. Classifies task (AI 0 vs Gig/Expert 1).<br>2. Performs <b>Automated AI Verification</b> of proof against requirements.        |

### 3. AI Proof Verification Engine (Key Innovation)

A critical feature judges will love is **Automated AI Verification**. When a worker submits a deliverable on the web portal, the backend calls Gemini / Groq with the initial task prompt and the worker's submitted proof to verify accuracy before releasing payment.

```
def verify_deliverable_with_ai(prompt: str, proof_text: str, proof_type: str):
    system_prompt = '''You are an AI Verification Auditor for Sarathi AI.
    Compare the User's Original Task against the Worker's Submitted Proof.
    Evaluate if the worker answered the core request accurately.
    Return JSON: {"is_verified": true|false, "confidence_score": 0-100, "feedback": "..."}'''

    user_payload = f"Task: {prompt}\nProof ({proof_type}): {proof_text}"
    # Calls Gemini Flash API -> returns JSON evaluation
    # If is_verified == true -> Auto-release UPI Escrow!
```

### 4. Dummy UPI Escrow State Machine

The ledger tracks tasks through a strict financial state machine in memory / database:

| State Code      | Trigger Event                        | Financial Action                                              |
|-----------------|--------------------------------------|---------------------------------------------------------------|
| UNPAID_QUOTE    | User sends task in Telegram          | AI calculates quote. No money moved.                          |
| ESCROW_HOLD     | User taps 'Approve & Pre-Auth' in TG | Pre-auth simulation: █X locked in user escrow ledger.         |
| CLAIMED         | Worker clicks 'Claim' on Web Portal  | Task locked to worker ID. Countdown timer starts.             |
| PROOF_SUBMITTED | Worker submits proof on Web          | AI Verification Engine checks proof accuracy.                 |
| SETTLED         | AI verifies proof (or User approves) | █X transferred from Escrow to Worker UPI ID. Tx ID generated. |

### 5. Teammate Setup & Task Allocation Checklist

Share this section with your teammates to divide and conquer:

#### A. 2-Minute Free Key Setup (For any teammate running the bot):

- **Telegram Token:** Message @BotFather on Telegram \$to\$ Send /newbot \$to\$ Copy token.
- **Gemini API Key:** Go to [aistudio.google.com](https://aistudio.google.com) \$to\$ Sign in \$to\$ Click Get API Key.

#### B. Task Division for Team Members:

| Team Role                      | Assigned Tasks & Deliverables                                                                                                |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Frontend Dev 1 (Marketing)     | Build Cascade-styled landing page at <code>&lt;/code&gt;</code> (Hero, Stack, Pricing, FAQ, 'Open App' link).                |
| Frontend Dev 2 (Worker Portal) | Build Worker Dashboard at <code>&lt;/code&gt;/worker&lt;/code&gt; (Job Feed, Claim button, Deliverable Upload modal).</code> |
| Backend Dev (API & Bot)        | Build Flask REST endpoints + Telegram Bot handler + AI Proof Verification function.                                          |
| Presenter / Pitcher            | Prepare SIH slide deck & practice live 6-step phone-to-website demo flow for judges.                                         |

**Summary for SIH Presentation:** Sarathi AI demonstrates a seamless real-world problem solution: no app downloads required for users, zero cost for free AI answers, pre-authorized UPI escrow for trust, and automated AI proof verification to protect both users and gig workers.